IoT dan Jaringan Sensor

Evaluasi IOT-MQTT



Dosen Pengampu :
Akhmad Hendriawan, ST, MT

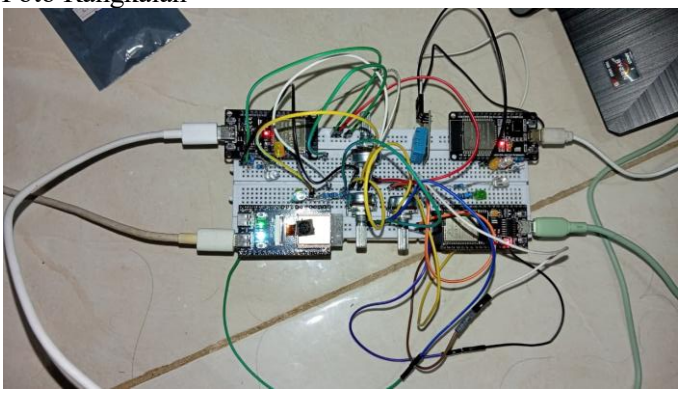
Nama Mahasiswa/NRP:

Inaayah Khairunnisa	2124600094 - Ruang 1
A'isy Ayyasy Mahmudi	2124600105 - Ruang 2
Reivan Haidar Arkana	2124600111 - Ruang 3
Wildan Rahmad Mulya	2124600113 - Ruang 4

PROGRAM STUDI SARJANA TERAPAN TEKNIK ELEKTRONIKA
DEPARTEMEN TEKNIK ELEKTRO
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA

2026

Foto Rangkaian



Tabel pin yang di pakai Ruangan 1 ,2 ,3,dan 4

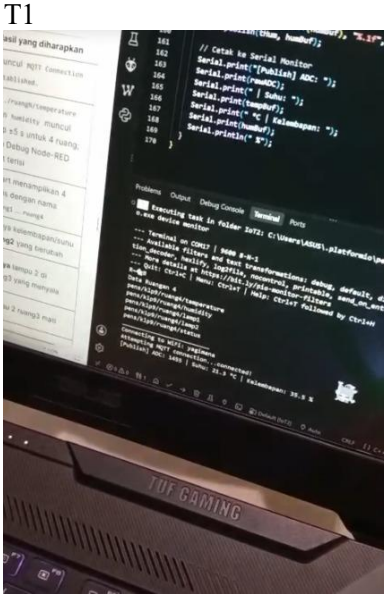
Ruang 1	Komponen	Potensiometer 10 kΩ	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO32	GPIO16	GPIO19
Ruang 2	Komponen	DHT	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO32	GPIO25	GPIO33
Ruang 3	Komponen	Potensiometer 10 kΩ	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO32	GPIO16	GPIO18
Ruang 4	Komponen	Potensiometer 10 kΩ	LED lampu 1 + resistor 220 Ω	LED lampu 2 + resistor 220 Ω
	Pin ESP32	GPIO6	GPIO16	GPIO18

Tabel topic final

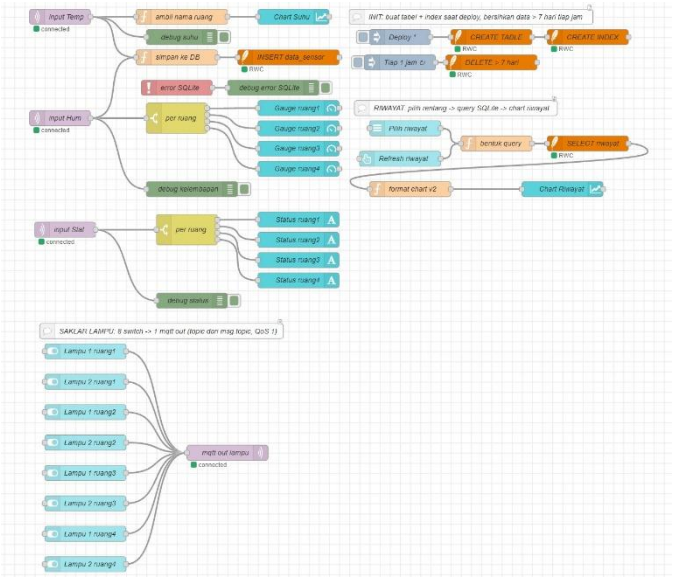
Topic	Arah	Payload	QoS	Retain
pens/klp9/ruangN/temperature	ESP32 → Node-RED	˚C, 1 Desimal, mis.28.5	0	Tidak
pens/klp9/ruangN/humidity	ESP32 → Node-RED	% 1 desimal, mis. 65.0	0	Tidak
pens/klp9/ruangN/lamp1	Node-RED → ESP32	1 = nyala 0 = mati	1	Tidak
pens/klp9/ruangN/lamp2	Node-RED → ESP32	1 = nyala 0 = mati	1	Tidak
pens/klp9/ruangN/status	LWT	Online atau offline	0	Ya

Hasil pengujian

T1



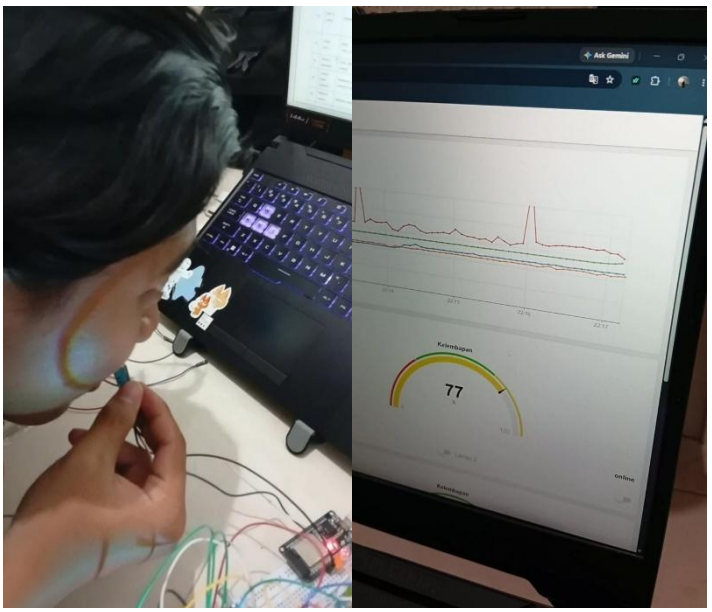
T2


Input Temp
connected
ambil nama ruang
Chart Suhu
debug suhu
simpan ke DB
INSERT data_sensor
RWC
error SQLite
debug error SQLite
Input Hum
connected
per ruang
Gauge ruang1
Gauge ruang2
Gauge ruang3
Gauge ruang4
debug kelembaban
INIT: buat tabel + index saat deploy, bersihkan data > 7 hari tiap jam
Deploy
CREATE TABLE
CREATE INDEX
Tiap 1 jam
DELETE > 7 hari
RIWAYAT: pilih rentang -> query SQLite -> chart riwayat
Pilih riwayat
bentuk query
SELECT riwayat
Refresh riwayat
format chart v2
Chart Riwayat
input Stat
connected
per ruang
Status ruang1
Status ruang2
Status ruang3
Status ruang4
debug status
SAKLAR LAMPU: 8 switch -> 1 mqtt out (topic dan msg topic, QoS 1)
Lampu 1 ruang1
Lampu 2 ruang1
Lampu 1 ruang2
Lampu 2 ruang2
Lampu 1 ruang3
Lampu 2 ruang3
Lampu 1 ruang4
Lampu 2 ruang4
mqtt out lampu
connected

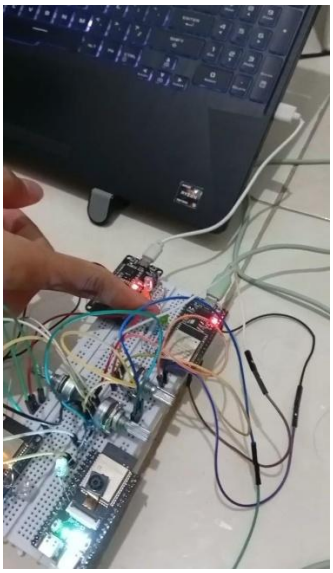
T3


37.1
Ruang 2
Kelembapan
45

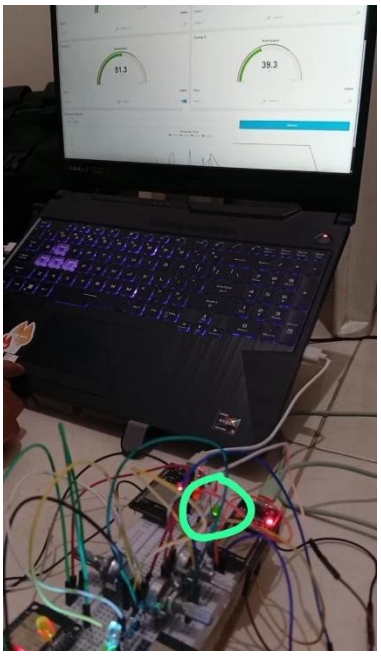
T4


Ask Gemini
Kelembapan
77
online

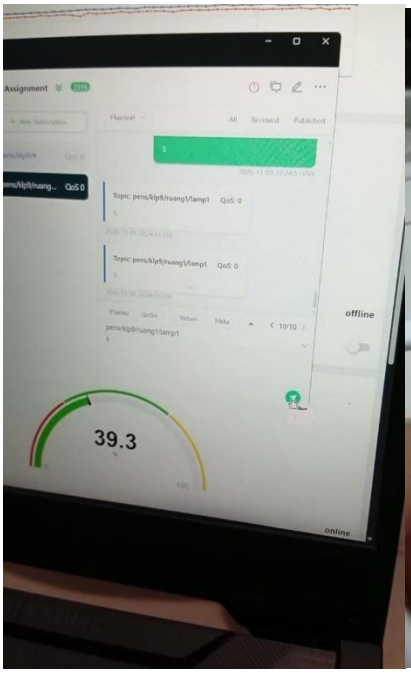
T5

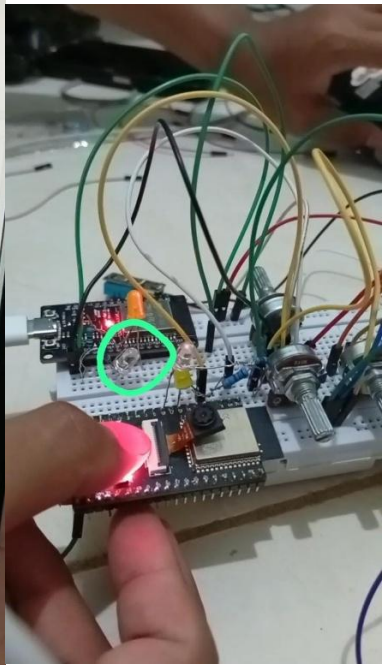


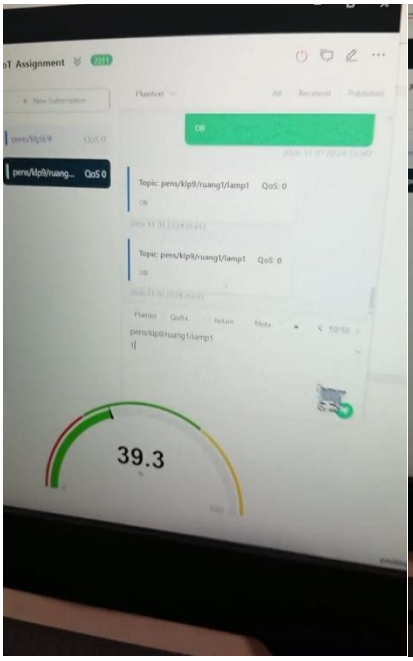
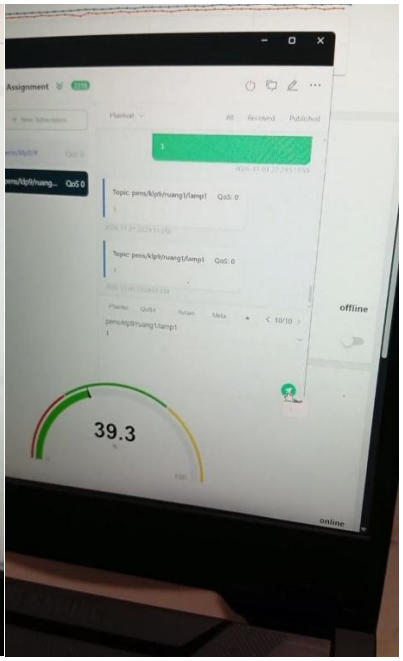
T6



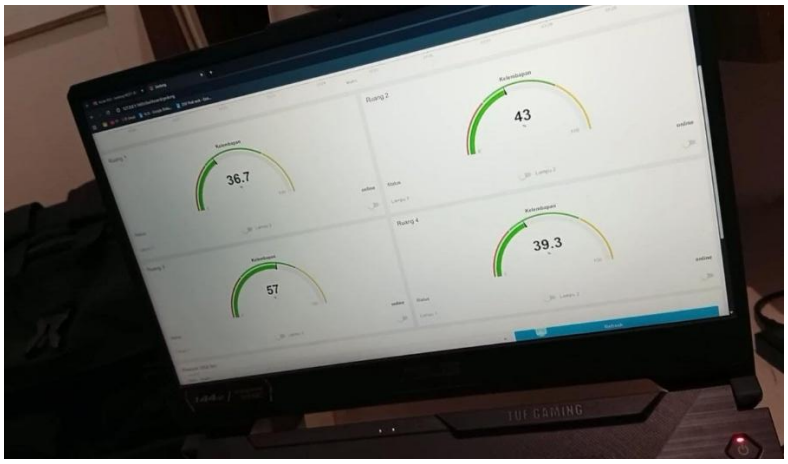
T7



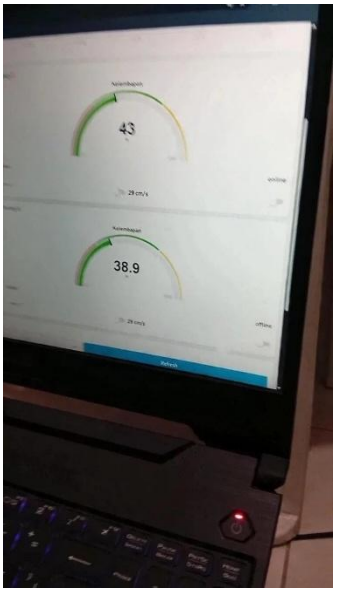



Topic: pens/klp9/ruang1/lamp1 QoS: 0
39.3

offline
39.3
online

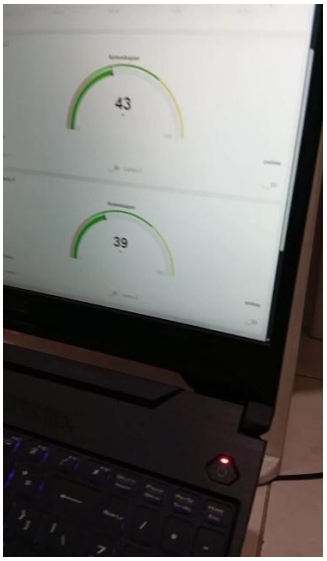
T8


36.7
43
57
39.3

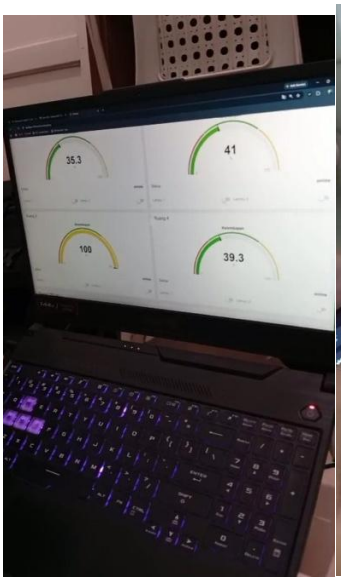
T9

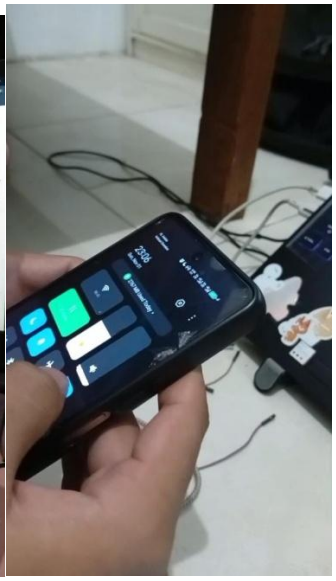

43
38.9

T10

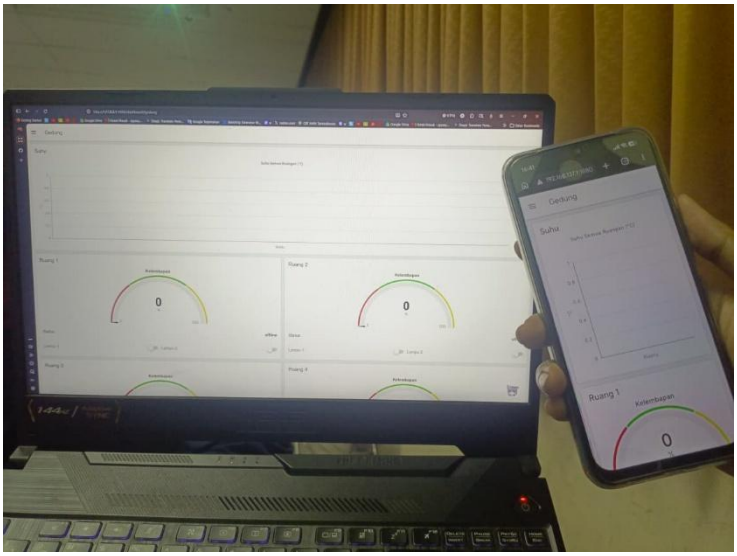


T11





T12



Kendala

Pada saat pengerjaan T12, solusinya saat menghubungkan HP ke laptop bukan hanya menyamakan koneksi sinyal tetapi juga IP pada servernya, jika tidak efeknya tidak akan terhubung pada HP